Rectangular mean observations with critical inverse moments

Lluis Eriksson

7 November 2026 — grouped supplement for the existing paper dossier

Abstract

We construct rectangular mean observations for every positive scalar coefficient on a finite periodic lattice that depends on one coordinate. At the critical inverse-moment exponent, the number of cells attains the target upper order, with the penalty coefficient equal to the coercivity threshold. The proof uses actual interval resistances, transverse averaging and a balanced-tree charge valid for odd lattice sizes. No bounded contrast or separation of weak sites is assumed. A thin weak layer separates this construction from certificates based only on the minimum coefficient. We also determine the exact asymptotic cell-cost phase diagram of strong Green-moment certificates on homogeneous coefficients, for arbitrary rectangular partitions and aspect ratios. At the critical exponent these certificates pay a logarithmic overhead although one actual mean suffices. An explicit coexact field refutes extension of the laminate argument by sampling a representative fibre. The unrestricted critical rectangular question remains open. These finite analytical results consolidate related positive and negative mechanisms; they do not establish a physical mass gap, formal verification or independent originality.

1 Problem and relation to the existing record

This grouped supplement belongs to the existing coarse-information paper dossier [1]. It combines the complementary research units184 and185 rather than proposing a deposit for each corollary. Let $D \geq 3$, $n \geq 2$, $X = (\mathbb{Z}/n\mathbb{Z})^D$, $N = n^D$ and $p = D/2$. All differences have unit spacing and all norms use counting measure. Set

$$\mathcal{E}_a(u) = \sum_{x \in X} \sum_{j=1}^{D} a(x)|u(x+e_j) - u(x)|^2, \qquad a(x) > 0.$$

For a disjoint rectangular partition $\mathcal{P}$ define the actual normalized arithmetic means $C_{\mathcal{P}}u = (\sqrt{|R|}\,\bar{u}_R)_{R \in \mathcal{P}}$. Our positive construction gives, simultaneously for all fields,

$$c\|u\|^2 \leq \mathcal{E}_a(u) + c\|C_{\mathcal{P}}u\|^2, \qquad c > 0. \tag{1}$$

For a fixed h-component field the output dimension is $h\#\mathcal{P}$ over the selected scalar field; complex dimension must be doubled when counted over the reals. Scalar estimates sum over components, and hence restrict to any actual coexact subspace without assuming an abstract surrogate.

The original question asks for a target-order partition under $\mu = \mathrm{mean}(a) \geq 1$, $1 \leq a \leq 2\mu$ and $\mathrm{mean}((\mu/a)^p) \leq M$, with cell count bounded by a constant depending on D, c, M times $\max(1, N/(1+\mu)^p)$. We solve here only the subclass $a(x) = \alpha(x_1)$. Lamination is an explicit retained hypothesis. The Green-moment results below concern the cost of a sufficient certificate, not a lower bound for every successful observation or for coexact rank.

2 A constructive theorem for arbitrary laminates

Put $d = D - 1$ and

$$r = (2/3)^d, \qquad B_D = \frac{2\,6^d}{1-r}.$$

Theorem 2.1. *For every positive sequence $\alpha(0), \dots, \alpha(n-1)$, set $a(x) = \alpha(x_1)$. There is a disjoint axis-aligned rectangular partition independent of u that satisfies* (1) *and*

$$\#\mathcal{P} \le 2^d + B_D(c/2)^p \sum_{x \in X} a(x)^{-p}. \tag{2}$$

If the original critical inverse-moment assumptions hold, its output rank is at most

$$h\big[2^d + B_D(c/2)^p 2^p M\big] \max\big(1, N/(1+\mu)^p\big).$$

The upper bound $a \le 2\mu$ is not needed for (2)*. Target order here means the requested upper order, not a matching lower bound for every individual coefficient.*

2.1 Actual local energy estimate

For an interval I of length m, write $R_I = \sum_{x \in I} \alpha(x)^{-1}$. Include its last starting vertex in this sum: although its longitudinal edge may leave the cut interval, its coefficient is needed for transverse averaging. Internal rectangle energies keep only edges within cut paths; periodic closing and cross-cell edges are discarded as nonnegative contributions.

Lemma 2.2. *If $R = I \times Z$ and every transverse side has length at most m, then*

$$\mathrm{Var}_R(u) := \sum_R |u - \bar{u}_R|^2 \le \frac{mR_I}{2} \mathcal{E}_{a,R}(u). \tag{3}$$

The constant $mR_I/2$ is an inverse-gap bound.

Proof. Let $v(y) = m^{-1} \sum_{x \in I} u(x,y)$. Orthogonal decomposition gives

$$\mathrm{Var}_R(u) = \sum_{y \in Z} \mathrm{Var}_I(u(\cdot, y)) + m\,\mathrm{Var}_Z(v).$$

The pairwise variance identity and weighted Cauchy along a path give $\mathrm{Var}_I \le (m-1) R_I \mathcal{E}_1/2 \le mR_I\mathcal{E}_1/2$. For an unweighted path of length ℓ, the same identity bounds the coefficient of its kth edge by $k(\ell - k) \le \ell^2/4$: use the path Cauchy bound with its length and then bound that length divided by ℓ by one. Tensorizing these path inequalities gives $\mathrm{Var}_Z(v) \le m^2 \sum_{y, j>1} |D_j v(y)|^2/4$; the tensorized constant is the largest coordinate constant, not their sum. Weighted Cauchy yields

$$|D_j v(y)|^2 \le \frac{R_I}{m^2} \sum_{x \in I} \alpha(x) |D_j u(x,y)|^2.$$

Thus $\mathrm{Var}_R \le (mR_I/2)\mathcal{E}_1 + (mR_I/4)\mathcal{E}_\perp$, proving the lemma. This is the step where a common coefficient sequence on all fibres is used. □

2.2 Partition algorithm and critical charge

Start from the interval $\{0, \dots, n-1\}$. Split a node of length $m \geq 2$ into consecutive lengths $\lfloor m/2 \rfloor, \lceil m/2 \rceil$ exactly when $mR_I > 2/c$. Accept equality. Never split a singleton. For each final interval of length m, partition every transverse coordinate into intervals of length m with a possible shorter last interval. Their products form an actual disjoint cover. A nonsingleton leaf satisfies $c \operatorname{Var}_R \leq \mathcal{E}_{a,R}$ by Lemma 2.2; singleton longitudinal leaves produce only singleton cells, with zero variance. Summing and using $\|u\|^2 = \sum_R \operatorname{Var}_R(u) + \sum_R |R||\bar{u}_R|^2$ proves (1).

For the count set $b(I) = \sum_I \alpha^{-p}$ and $w(I) = m^{-d}$. Hölder gives $R_I \leq m^{1-1/p} b(I)^{1/p}$. Every split therefore satisfies

$$w(I) < (c/2)^p b(I), \qquad d = 2p - 1. \tag{4}$$

Let T consist of internal nodes with no internal children. These intervals are disjoint. Assign each internal node to a terminal internal descendant. Each child has size at most $2/3$ of its parent, also when sizes are odd. Nodes assigned to a fixed terminal node lie on its ancestor path at distinct depths, so their weights sum to at most $w(J)/(1-r)$. Consequently

$$\sum_{I \text{ internal}} w(I) \leq \frac{(c/2)^p}{1-r} \sum_{x_1} \alpha(x_1)^{-p}.$$

Each child also has size at least $m/3$. Telescoping leaf weights over all splits gives

$$\sum_{I \text{ leaf}} w(I) \leq n^{-d} + 2\,3^d \sum_{I \text{ internal}} w(I).$$

The exact transverse cost of a leaf is $\lceil n/m \rceil^d \leq (2n/m)^d$. Multiply the last display by $(2n)^d$ and use $\sum_X a^{-p} = n^d \sum_{x_1} \alpha^{-p}$ to obtain (2). Finally $\sum_X a^{-p} \leq MN/\mu^p$ and $\mu^{-p} \leq 2^p (1+\mu)^{-p}$ prove the stated target bound. This completes the proof of Theorem 2.1 without a logarithmic charge at each tree level.

2.3 A thin weak layer beyond minimum certificates

Take integer $t \geq 2$, $n = t^D$, $\mu = n^2$, $b = n^{2-2/D}$, and

$$\alpha(0) = b, \qquad \alpha(i) = H = \frac{n\mu - b}{n-1} \quad (i > 0).$$

Then $1 \leq \alpha \leq 2\mu$, its exact mean is μ, and

$$\operatorname{mean}((\mu/a)^p) = 1 + (1 - 1/n)(\mu/H)^p < 2. \tag{5}$$

The uniform bound for this unbounded sequence is $M = 2$; the moments tend to two, so a single fixed $M < 2$ is not asserted. Moreover

$$nR_{\text{root}} = n^{-1+2/D} + \frac{(n-1)^2}{n^2 - b/n} < 2.$$

Thus at $c = 1$ the algorithm accepts the root and uses one mean. Yet $b \to \infty$ and $b/\mu \to 0$, excluding every fixed coefficient-gap condition $a \leq B$ or $a \geq \theta\mu$ for fixed $B, \theta > 0$ eventually.

Consider instead any partition whose cells pass the minimum-only sufficient test $(\min_R a)\lambda_1^0(R) \geq 1$, where λ_1^0 is the actual unweighted internal gap. For a singleton the

mean-zero condition is vacuous. Every cell meeting the weak layer has minimum b. A linear path test gives $\lambda_1^0 \le 12/\ell^2$ in a cut coordinate of length $\ell \ge 2$; a Fourier test gives $\lambda_1^0 < 40/\ell^2$ in a full cycle coordinate. Thus each transverse side of such a cell is at most $\sqrt{40b}$. Covering the n^d weak-layer vertices forces

$$\#\mathcal{P} \ge 40^{-d/2} n^{d/D}.$$

This is a limitation of that certificate, since the preceding one-cell construction works on the same coefficients.

3 The cell cost of strong Green moments

Now take homogeneous $a = \mu = n^2$. These original coefficient data have $M = 1$ and target order one. On a rectangle R use cut coordinate paths, or retain a periodic edge only when its coordinate spans the full period. Let L_R be its actual weighted Laplacian and $G_R = L_R^\dagger$, zero on the constant kernel. For $s > 0$ let

$$T_s(R) = \operatorname{tr}(G_R^s) = \sum_{\lambda_j(L_R)>0} \lambda_j(L_R)^{-s}.$$

For a singleton set $T_s = 0$. The strong certificate $T_s(R) \le 1$ indeed implies $\operatorname{Var}_R(u) \le \mathcal{E}_{a,R}(u)$, because its first positive inverse eigenvalue raised to s is one summand. Define $q_s(n)$ as the minimum number of cells in any disjoint rectangular partition passing this test. No hierarchy, aspect ratio or placement restriction is imposed.

Theorem 3.1. *For fixed $D \ge 3$ and $s > 0$,*

$$q_s(n) = \begin{cases} \Theta_{D,s}(n^{D-2s}), & 0 < s < D/2, \\ \Theta_D(\log n), & s = D/2, \\ \Theta_{D,s}(1), & s > D/2. \end{cases}$$

The actual gap and a weak critical counting certificate both allow one root cell on this family. This phase diagram concerns sufficient certificate cost, not necessary observation rank.

3.1 Finite spectra and all-partition lower bounds

The cut path of length ℓ has eigenvectors $\cos(\pi k(x + 1/2)/\ell)$ and eigenvalues $4\sin^2(\pi k/(2\ell))$, $0 \le k < \ell$. The endpoint identity follows from cosine addition and $\sin(\pi k) = 0$. Distinct eigenvalues give a full orthogonal basis. A full cycle has the usual Fourier eigenvalues $4\sin^2(\pi k/n)$. At $n = 2$ retain both positive-start periodic edges; its nonzero eigenvalue is four, while the cut path eigenvalue is two. Tensor products give the cut-rectangle spectrum

$$\lambda_k = 4n^2 \sum_{i=1}^{D} \sin^2 \frac{\pi k_i}{2\ell_i}, \qquad 0 \le k_i < \ell_i, \quad k \ne 0. \tag{6}$$

Adding full-period edges increases ordered eigenvalues by min-max and keeps the constant kernel. Comparing eigenvalues suffices for every $s > 0$; we do not assume an operator-power order valid for arbitrary powers.

Every such internal graph has $\lambda_{\max} \le 4Dn^2$. Hence, with $V = |R|$, $T_s(R) \ge (V-1)/(4Dn^2)^s$. A passing partition satisfies

$$q \ge \frac{n^D}{1 + (4Dn^2)^s}. \tag{7}$$

This proves the subcritical lower bound for arbitrary shapes.

For the endpoint set $p = D/2$, $m = \min_i \ell_i$, $J(m) = \max(0, \lfloor \log_2 m \rfloor - 2)$ and $A_D = 2^{-D}(256D)^{-\lceil D/2 \rceil}$. We claim

$$T_p(R) \ge A_D (V/n^D) J(m). \tag{8}$$

It is vacuous if $m < 8$. Otherwise for $j = 0, \dots, \lfloor \log_2 m \rfloor - 3$ put $t_j = 2^{j+1}/m$ and select the disjoint mode blocks $\lceil t_j \ell_i \rceil \le k_i < \lceil 2t_j \ell_i \rceil$ for every i. Their sizes are at least $V t_j^D / 2^D$: for $x = t_j \ell_i \ge 2$, $\lceil 2x \rceil - \lceil x \rceil \ge x/2$. All selected frequencies have $k_i/\ell_i < 2t_j \le 1/2$ and satisfy $\lambda_k < 256 D n^2 t_j^2$, under both path and retained-cycle spectra. Their contribution is at least $A_D V/n^D$ per block. This proves (8), including arbitrary aspect ratios.

In a partition with q cells, those of volume at least $N/(2q)$ cover at least half of X. Each of their sides is at least $n/(2q)$. Summing (8) over these cells and using $T_p \le 1$ gives

$$q \ge \frac{A_D}{2} \max\{0, \lfloor \log_2(n/(2q)) \rfloor - 2\}.$$

For $n \ge 2^{16}$ this implies $q \ge (A_D/8) \log_2 n$. Indeed if $q \ge \sqrt{n}$ this is immediate; otherwise the expression in braces is at least $(\log_2 n)/2 - 4 \ge (\log_2 n)/4$.

3.2 Matching actual meshes

Use a Cartesian mesh with all side lengths at most an integer $M \le n$, allowing shorter last intervals. From (6) and $\sin t \ge 2t/\pi$ on $[0, \pi/2]$, group modes by $j = \max_i k_i$ to get

$$T_s(R) \le D 2^{D-1} 4^{-s} (M/n)^{2s} \sum_{j=1}^{M-1} j^{D-1-2s}. \tag{9}$$

There are at most $D(j+1)^{D-1} \le D 2^{D-1} j^{D-1}$ modes in the group. When $2s < D$, the sum is at most a constant depending on D, s times M^{D-2s}. Choose $M = \max(1, \lfloor \varepsilon n^{2s/D} \rfloor)$ with fixed sufficiently small $\varepsilon > 0$. Then every trace passes, and the cell count is $O_{D,s}(n^{D-2s})$. Floor-zero cases use singletons. When $2s = D$, the sum is at most $1 + \log M$. Choosing $M = \max(1, \lfloor \varepsilon n/(1+\log n)^{1/D} \rfloor)$ gives $O_D(1 + \log n)$ cells. When $2s > D$, the sum is at most $1 + (2s-D)^{-1}$; choose $M = \max(1, \lfloor \varepsilon n \rfloor)$ to get a constant cell count. Retained full-period edges only decrease the traces. These constructions and the lower bounds prove Theorem 3.1.

3.3 Successful weak endpoint and exact finite distinctions

For the cut root, $\lambda_k \ge 4|k|^2$, so its true positive gap is at least four. Define

$$W_p = \sup_{\lambda > 0} \lambda^{-p} \#\{0 < \lambda_j(L_X) < \lambda\}.$$

There are no such eigenvalues for $\lambda < 4$; for $\lambda \ge 4$ their number is at most $(1 + \sqrt{\lambda}/2)^D \le \lambda^p$. Thus $W_p \le 1$. The same holds after adding periodic edges. Integrality of this counting function excludes any eigenvalue below one, so this weak criterion really certifies the successful one-mean

inequality. The weak/strong ideal distinction is classical [5]; its use does not solve the original coefficient-to-rectangle construction.

For $D = 4, n = 4$, the periodic coordinate spectrum at $a = 16$ is $0, 32, 64, 32$. The root eigenvalues $32m$ have multiplicities $\binom{8}{m}$. Its gap is32, but $T_1(X) \geq 56/96+70/128 = 217/192 > 1$; sixteen cut length-two boxes instead have $T_1 = 103/384$ each. At the critical endpoint an exact separate example takes $n = 2$, $\mu = 4$, $c = 16$ and test $16^p T_p(R) \leq 1$. A rectangle with r full coordinates has spectrum $16m$ with multiplicity $\binom{r}{m}$, so this test passes exactly when $r \leq 1$. Therefore its minimum cell count is 2^{D-1}, whereas the actual one-mean inequality works with $K = c = 16$. For $D = 4$ the rejected root value is $865/144$. Changing a fixed c changes the constants in the phase diagram, not its orders. The ordinary identity $\sum_{x<y} R_{\mathrm{eff}}(x, y) = |R| \operatorname{tr}(G_R)$ follows by expanding $(e_x - e_y)^T G_R(e_x - e_y)$ and using $G_R \mathbf{1} = 0$; it only identifies the all-pairs resistance sum with the $s = 1$ test, not every resistance method.

4 A coexact falsifier for representative fibres

Lamination cannot be replaced by sampling one favourable fibre. In $D = 3, n = 4$ let $c = 15/2$ and set $a = 1$ on $\{0, 1, 2\}^3$, $a = 64$ elsewhere. Its exact mean is $\mu = 2395/64$, $1 \leq a \leq 2\mu$, and

$$\operatorname{mean}((\mu/a)^{3/2}) < \frac{27}{64}\frac{1178}{5} + \frac{37}{64} = \frac{31991}{320} < 100.$$

Here $\mu < 38$ and $\sqrt{\mu} < 31/5$. The fibre through $(3, 3)$ in the transverse coordinates has all coefficients64 and would pass $nR = 1/4 < 2/c = 4/15$, leading to a false one-cell conclusion.

Let $G = \delta_{(1,1,1)}$ and use backward differences to define $U_1 = D_2^- G$, $U_2 = -D_1^- G$, $U_3 = 0$. This is the codifferential of an actual antisymmetric plaquette potential, up to its global sign convention; backward divergence vanishes by commuting differences. Both nonzero components have zero mean and squared norm two, so $\|U\|^2 = 4$. An adjacent opposite delta pair has forward-gradient energy six in its own direction and four in each other direction. The total for U is28. Every nonzero forward-gradient starting vertex lies in the weak cube, so the actual weighted energy is also28. Consequently $\mathcal{E}_a(U) = 28 < 30 = c\|U\|^2$, while the global mean penalty vanishes. This refutes the representative-fibre mechanism even on the actual coexact domain. A different partition remains allowed by the original problem, so that problem has not been refuted.

5 Literature comparison and verification boundaries

Weighted path Cauchy estimates and weighted Hardy theory are classical; Muckenhoupt [2] is relevant context. Adaptive tree approximation has an extensive literature, including Binev–DeVore [3]. Our tree charge is proved explicitly rather than imported from a near-best approximation theorem. Publisher/author metadata for those two references were checked; unavailable full-text retrievals are not represented as complete comparisons.

Nochetto–Otárola–Salgado [4], Lemma3.1 and Theorem5.7, give constructive weighted Poincaré and anisotropic rectangular interpolation estimates in Muckenhoupt-weighted Sobolev spaces. Their constants depend on the weight characteristic and mesh geometry. Their inspected results are not the finite counting-norm observation theorem proved here, nor a proof of its uniform constants from only a global inverse moment. Conversely this distinction is not proof of originality: broader weighted approximation and adaptive-partition literature still requires

comparison. In particular no claim is made to invent weighted Poincaré inequalities, spectral counting, adaptive trees, or weak Schatten ideals. The weak/strong distinction is separately attributed to the established spectral literature [5].

The universal mathematical arguments above are analytical, not Lean verified. Source units184 and185 have exact rational finite checks at pinned Git commits. Unit184 has four matching normal/optimized payloads from two fresh Colab clones, six rejecting mutations, five critical matrix controls and six mode-block cases. Unit185 has four matching payloads from two fresh clones, six global matrix controls,72 balanced-tree cases, five weak-layer instances and six rejected intentional errors. It checks the actual coexact energy28 versus30. These are selected finite tests, not an all-size computation or independent replication. Full stdout, stderr, actual exit codes, hashes, failures and resource records are in the source evidence directories. No scientific computation ran on Windows.

5.1 Model assessments and review context

This exact supplement PDF has no inherited numerical grade or stars. It is a draft pending its own declared exact-PDF assessment, not a public AIRR release, human peer-reviewed paper or editorial acceptance. Public version8 of the existing dossier has a separate exact-PDF AI assessment; its score does not transfer to these bytes.

The proof of unit185 was read once in full in an adversarial source audit; the auditor found no blocking defect in the laminate construction or critical charge and recorded the uniform $M = 2$ distinction and dimension, singleton and target-order qualifications incorporated here. This audit has no numeric score. Interface: OpenAI Codex, configured gpt-6-astra; the actual served backend and reasoning setting were unexposed. Shared account, provider and historical chat context mean the audit is neither blind nor organizationally independent. The full report is retained, including objections; it is not replaced by this summary. It did not review this newly typeset exact PDF. No assessment was rerun to improve a grade, and no report is represented as human peer review.

5.2 Reproducibility and remaining question

The source proof for185 has SHA256

`cee3556c1ff9dd99a69b3d62b20f65288189`
`26cca2734b2728b51ce9e4fdd407`

The corresponding executed commit is `dfc2205dcfb8642bf5639c8433e4bdbff38075a3`; source184 was executed at `bedf55c593994aeba2cf47454c2f608870ecb871`. This supplement's source and rendered-file hashes belong in its separate build manifest; a PDF cannot contain its own final hash. The research evidence is preserved at the pinned programme repository [6].

The sole unrestricted critical rectangular question remains open. A proposed extension must derive genuine transverse or all-fibre control from its coefficient hypotheses; one favourable fibre is insufficient. The strong Green test alone is abandoned at and below the critical exponent as a uniform target-order mechanism by Theorem 3.1. These findings do not reopen previously stopped routes, change PR29's 20/41 and TermSource0 baseline, or identify a finite sequence bound with a continuum or interacting Yang–Mills mass gap.

References

[1] L. Eriksson, *Coarse information and finite rank obstructions for flat lattice Hodge forms*, existing AIRR dossier ARR-2026-7EPM3V3DP796BBRR, public version8. `https://airr.science/papers/ARR-2026-7EPM3V3DP796BBRR/versions/v8/`. Working paper with AI assessment, not editorial acceptance.

[2] B. Muckenhoupt, *Hardy's inequality with weights*, Studia Mathematica44 (1972),31–38. `https://doi.org/10.4064/sm-44-1-31-38`.

[3] P. Binev and R. DeVore, *Fast computation in adaptive tree approximation*, Numerische Mathematik97 (2004),193–217. `https://doi.org/10.1007/s00211-003-0493-6`.

[4] R. H. Nochetto, E. Otárola and A. J. Salgado, *Piecewise polynomial interpolation in Muckenhoupt weighted Sobolev spaces and applications*, arXiv:1402.1916v2 (2014), Lemma3.1, Section5.2 and Theorem5.7. `https://arxiv.org/abs/1402.1916v2`.

[5] R. L. Frank, *Cwikel's theorem and the CLR inequality*, arXiv:1206.3325v1 (2012), Section2. `https://arxiv.org/abs/1206.3325v1`.

[6] L. Eriksson programme research evidence, units184–185, pinned at commit506004b3e7251aa881b08dde1a1bb14aff33e76f. Source proofs and complete finite evidence.